IILM UNIVERSITY
Gurugram | Greater Noida

INTERNSHIP PRESENTATION

AI/ML Internship

PRANAV MODI
Roll No. 25scs1003002600 • Section 2CSE26

04 WEEKS

30 JULY — 31 AUGUST 2026

CODEC
Technologies • AI/ML Internship

Submitted to IILM University, Greater Noida, Uttar Pradesh

INTERNSHIP AT A GLANCE

Four weeks connected AI/ML concepts with practice

The programme moves from AI and Python foundations to machine learning, neural networks, advanced applications and project work.

01

Certificate

04

Learning weeks

02

Projects required

Host: Codec Technologies • Role: AI Intern • July–August 2026

STUDENT PROFILE

Pranav Modi

Roll number
25scs1003002600

Section
2CSE26

Institution
IILM University, Greater Noida

LEARNING ROADMAP

Five stages connect AI concepts with practice

The four-week curriculum covers AI and Python, machine learning, neural networks and advanced applications, followed by project work.

WEEK 1	WEEK 2	WEEK 3	WEEK 4	PROJECTS
01	02	03	04	05
AI & Python	**ML basics**	**Neural networks**	**AI applications**	**Project delivery**
Concepts and coding	Learning from data	Layers and learning	Text, images, prediction	Build, test, document

LEARNING LOGIC

Foundations → ML → neural networks → applications → projects

AI and Python establish the learning foundation

illustrative_python.py

```
values = [2, 4, 6, 8]
mean = sum(values) / len(values)
centered = []
for value in values:
    centered.append(value - mean)
print(centered)
```

Foundation concepts

- Understand AI tasks such as recognition and prediction
- Relate machine learning to patterns learned from data
- Use Python to express clear, reusable processing steps
- Check inputs and explain the meaning of each output

EXAMPLE

The sample centers numeric values around their mean; it is a learning example, not a project result.

Machine learning links data to useful predictions

A learning task connects input features, a suitable model and an evaluation method that reflects the intended output.

INPUT — Relevant features and examples

TASK — Prediction or classification

MODEL — Method suited to the problem

CHECK — Evaluation on unseen data

conceptual_ml_workflow

```
Define the input and target
Separate training and test data
Choose a suitable model
Fit using training examples
Predict on unseen examples
Evaluate and explain errors
```

PRINCIPLE • Evaluate the model against the task, not training performance alone

Neural networks learn representations from data

conceptual_network_flow

```
Input features
    ↓
Hidden layers transform information
    ↓
Output: a prediction or class

Training adjusts model parameters
```

Learning focus

- Understand inputs, layers and model outputs
- Relate training examples to parameter learning
- Connect CNNs with image recognition tasks
- Assess predictions and inspect errors

PROJECT OPTIONS • Handwritten digit recognition and fruit image classification

AI applications use different inputs and outputs

The project catalogue spans numerical prediction, language processing, image recognition, recommendations and audio.

01 **PREDICTION**
Stock, churn and weather tasks

02 **LANGUAGE**
Sentiment, spam and chatbots

03 **VISION**
Digits and fruit image classes

04 **AUDIO**
Speech-to-text transcription

illustrative_input_output_pairs

```
Text → sentiment label
Image → digit or fruit class
Historical data → predicted value
Ratings → movie suggestions
Audio → transcribed text
```

KEY IDEA • The input and intended output guide the choice of method

Two selected projects apply the learning

The platform requires any two listed projects, publication on GitHub or LinkedIn, and sharing the link with the offer letter.

01 **DEFINE**
Clarify user need

02 **PROCESS**
Prepare task inputs

03 **MODEL**
Select an approach

04 **EVALUATE**
Inspect outputs

05 **DOCUMENT**
Results and usage

PROJECT RECORD • Add your actual selected titles and results on slide 12

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The project brief offers task-specific tools

FOUNDATION	**Core Python**	Python foundations • structured problem-solving
MODELS	**Prediction**	Linear regression • Naive Bayes • SVM
VISION	**Image tasks**	CNNs • MNIST for handwritten digit recognition
LANGUAGE	**Text & audio**	NLTK / spaCy • SpeechRecognition

These are programme options; the final tool list depends on the two selected projects.

SOLUTION WORKFLOW

A repeatable workflow links problem to evidence

This conceptual AI/ML workflow connects a clear objective with data preparation, modelling, evaluation and documented output.

1 **DEFINE** Clarify goal and data

2 **PREPARE** Clean and split the data

3 **MODEL** Choose and train a model

4 **EVALUATE** Assess unseen examples

5 **SHARE** Document and publish

✓ **Readability**

✓ **Clear objective**

✓ **Data quality**

✓ **Reliability**

✓ **Documentation**

PRINCIPLE • Define, prepare, model, evaluate and explain

AI/ML work requires careful data and evaluation

The following general risks explain why data quality, suitable methods and honest evaluation matter in an AI/ML project.

CHALLENGE	TECHNICAL RESPONSE	INTENDED EFFECT
INPUTS VARY	Validate types, formats and missing values	**Fewer avoidable processing failures**
MODEL OVERFITS	Evaluate on examples excluded from training	**A clearer view of generalization**
TASK MISMATCH	Match the method and checks to the objective	**More meaningful model assessment**
RESULTS UNCLEAR	Report outputs, errors and limitations	**Transparent, explainable reporting**

These are general project considerations; no numerical model-performance claims are made.

PROJECT RECORDS

Document the two completed projects

Selected project briefs: Stock Price Predictor and Movie Recommendation System.

01
Project 1: Stock Price Predictor
Objective: Predict stock prices using historical data.

02
Project 1 Method: Linear Regression
Use historical stock data, split train/test data, train the model and predict prices.

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03
Project 2: Movie Recommendation System
Objective: Suggest movies based on user ratings or genres.

04
Project 2 Method: Filtering
Use ratings or genres to match similar movies and recommend top suggestions.

05
Selected project work
Stock prediction method • Recommendation filtering method • AI/ML application practice

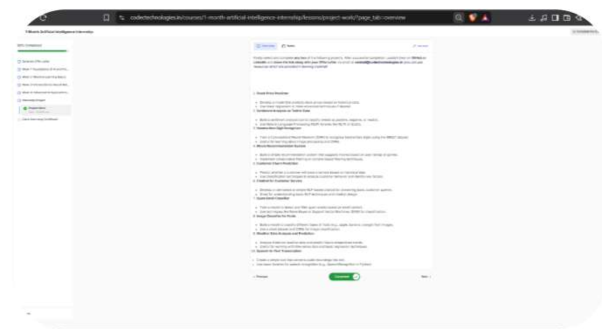
PROJECTS

Attached records confirm internship completion

01

COMPLETION CERTIFICATE

The certificate records Pranav Modi's one-month Artificial Intelligence internship at Codec Technologies.



SKILLS GAINED

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The internship connects technical and practical learning

TECHNICAL DEPTH

Understand AI/ML work from inputs to evidence

- Explain AI and machine learning concepts
- Connect data preparation with modelling
- Understand neural-network applications
- Match tools and checks to the task
- Document methods, results and limitations

PROFESSIONAL HABITS

01 Progressive problem solving
Break requirements into clear, testable steps.

02 Self-paced learning
Follow a structured sequence of learning topics.

03 Quality awareness
Check data, assess outputs and explain limitations.

04 Professional communication
Document work and prepare it for professional sharing.

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CONCLUSION

Connecting AI/ML concepts with practical applications

The internship links AI and Python foundations, machine learning, neural networks and application-oriented project work in one learning journey.

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NEXT APPLICATION

01 **Build further AI/ML projects**

02 **Strengthen model evaluation**

03 **Publish documented work**

Acknowledgement

With gratitude to IILM University and Codec Technologies for enabling this practical learning experience.